The ball-bearing motor and the Space Power Generator

A measurement registry for the Huber motor, and whether bearing torque can explain the SPG's figures — register entries H13 and H14

kronspice simulation report · 2026-11-07

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Summary

A current through a rotating ball bearing drives the shaft with a torque that scales as I^2, aids the rotation and is zero at rest (the Huber effect). kronspice carries it as the P element (device/bearing.go). This example holds the public record of what has been measured about such motors, the fitting code that would set the element's two coefficients, and one question: could bearing torque, aiding the drive motor, account for the published figures of Tewari's Space Power Generator (SPG, register H13)? Marinov's claim of heat beyond the electrical input is register entry H14.

- **No coefficients are invented.** No measured (I, ω, T) or back-EMF table is open access, so the dataset list is empty and the P element refuses to run without fitted κ and ω_0. The program lists the nine papers whose tables would fix them.
- **The SPG's torque gap is 24.8 N·m (7.01 kW):** the claimed output needs 41.5 N·m at the shaft, the drive motor supplied 16.7 N·m.
- **EM mode (the measured mode) leaves 1.37 V:** the bearings' back-EMF takes 2.04 V of the 3.41 V no-load EMF, so the rest of the loop gets 4.71 kW, not 11.72 kW. That is exactly the loaded-EMF bound of the SPG ledger.
- **Thermal mode is excluded:** at the measured efficiency η = 0.10 the bearings would need 70 kW of electrical input, 20.4 V across them, more than the 3.41 V the machine makes.
- **The engine run agrees:** whatever the contact resistance (1–100 μΩ), the load gets at most (E_0 − back-EMF)·I, with ledger residual 0.

What is being tested

H13 — Tewari's Space Power Generator

The SPG test (Tewari, *Universal Principles of Space and Matter*, pp. 159–160) reports 2700 r/min, 3438 A, a no-load EMF of 3.41 V and a drive motor rising from 2943 W to 8620 W at η = 0.83. **Standard prediction:** the generator's output is paid for by its shaft; the loaded EMF is the unmeasured quantity on which the claim rests (predicted ≤ 1.37 V, assumed 3.41 V). This example asks whether a known, measured mechanism, the bearing motor, could close the gap between the shaft work supplied and the electrical output claimed.

H14 — Marinov's excess heat

Marinov (*Electronics & Wireless World*, April 1989) reports, from a calorimeter, about 8 % more heat with rotation than without. This is a **claim** with no data shown, and it is not used in any calculation. **Standard prediction**, given by the P element's ledger: heat = V·I − ω·T.

The registry

Every number is a record with its source and a status: *measured* (reported from an experiment), *estimate* (no instrument named) or *claim* (asserted, no data shown). Only measured values enter a calculation.

finding	source	status
Steady currents 43.5–70.15 A; self-rotation at high current, small efficiency; a drive motor needed at low current	Moyssides & Hatzikonstantinou, IEEE Trans. Magn. 26, 1274 (1990), abstract	measured

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finding	source	status
Back-EMF that depends on shaft speed; no generator action	same	measured
Self-rotation 70–140 A, aided 18–140 A (two discs)	Moyssides 1992, via Lauterbach, Soong & Abbott (2001)	measured, secondary
Torque proportional to I²	Choo, Soong & Abbott, Proc. SPIE 5649 (2005), per McDonald (2020)	measured
Mechanical output / electrical input = 0.10	Marinov, Electronics & Wireless World, April 1989 (calorimeter)	measured, data unpublished
Heat 8 % higher with rotation than without	Marinov 1989	claim (H14), not used
1000 r/min (Milroy 1959); "at least 1500 rev/min" (Ogden 1989)	Milroy, J. Appl. Mech. 34, 525 (1967); Ogden 1989	measured / estimate
Needs a hand spin to start; thermal effects start it, the Lorentz force cannot	Harrison (electricstuff); McDonald (2020) fn. 6	measured
Bearing temperature 80 °C at 30 A	Univ. Adelaide 2022	measured

The model

The P element's torque law is $T = \kappa \cdot I^2 \cdot \tanh(\omega/\omega_0)$, where I is the current through the bearing, ω the shaft speed, κ a torque coefficient and ω_0 a speed scale. It runs in one of two modes: in EM mode the element has a back-EMF E with $E \cdot I = \omega \cdot T$, so electrical power is converted to shaft work; in thermal mode there is no back-EMF: the element is a heat engine on the contact heating, with shaft work at most η times that heat. FitTorque fits κ and ω_0 to a measured (I, ω, T) table and FitBackEMF to a measured back-EMF table; in EM mode the two fits must agree.

Because no such table is open access, the dataset list is empty and the element refuses to run without fitted KAPPA and W0. The program lists the nine papers whose tables would fix the coefficients: Gruenberg 1978, Moyssides 1990 and 1997, Watson 1993 and 1999, Choo 2005, Shen 1999, Polivanov 1973 and Huber 1959. TestFitRecovers checks the fitting code on synthetic points (κ and ω_0 recovered to 10^{-6}), not on a bearing.

Results: could bearing torque explain the SPG?

All inputs are measured: the SPG test figures above and the bearing efficiency η = 0.10.

quantity	value
shaft torque the claimed $E_0 \cdot I$ needs	41.5 N·m
shaft torque the drive motor supplied	16.7 N·m
gap a bearing torque would have to fill	24.8 N·m, 7.01 kW
$\kappa \cdot \psi(\omega)$ needed if all 3438 A cross the bearings	≥ 2.1×10^{-6} N·m/A²

- **EM mode** (the measured mode: a back-EMF exists). The bearings then drop 2.04 V of the 3.41 V, and the rest of the loop gets 4.71 kW, not 11.72 kW. The 1.37 V left is exactly the loaded-EMF bound of the SPG ledger: a bearing motor moves shaft work around and adds none.
- **Thermal mode** at the measured η = 0.10: the bearings would need 70 kW of electrical input, 20.4 V across them, more than the 3.41 V the machine makes. Excluded.

Engine run (SimulateSPGLoop)

One circuit holds the no-load EMF, the P element at the SPG's speed with the derived κψ, and the rest of the loop sized for 3438 A. The unknown bearing contact resistance R_b is swept:

R_b	load	bearing shaft work	$E_0 \cdot I$	ledger residual
1 μΩ	4.70 kW	7.01 kW	11.72 kW	0
10 μΩ	4.59 kW	7.01 kW	11.72 kW	0
100 μΩ	3.53 kW	7.01 kW	11.72 kW	0

The load never gets more than (E_0 − back-EMF)·I, whatever R_b. The exported netlist and results sheet (export/bearing_motor_spg_loop_results.md) are solved by MNA only; Kron's torn solution is shown for linear circuits.

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Validation

- main_test.go: fit recovery on synthetic data, every registry entry sourced, the SPG bound (24.8 N·m, 1.37 V) and the swept engine run.
- Ledger residual 0 at every contact resistance in the sweep.

Limits

- κ and ω_0 are not known for any real bearing; the analysis uses only the product $\kappa \cdot \psi(\omega)$ the SPG would need, and asks whether energy could balance, not whether a bearing could produce that torque.
- The SPG figures are Tewari's published numbers; the loaded EMF was not measured.
- Marinov's 8 % excess heat is a claim with no data and is not used.

Status and next tests

H13 stays *untested*: the SPG test rests on an unmeasured loaded EMF (≤ 1.37 V predicted, 3.41 V assumed); bearing torque could close the 24.8 N·m gap only in EM mode, whose back-EMF leaves the same 1.37 V, and is excluded in thermal mode at η ≈ 0.10. **H14** stays *untested*: no data published; the standard prediction is the P element's ledger, heat = V·I − ω·T.

Decisive bench tests: insulate the SPG bearings (ceramic balls or insulated housings) and repeat the loaded run; measure the voltage across each bearing under load; put a torque transducer between the drive motor and the rotor. For H14, a calorimetric repeat of Marinov's run with the ledger above as the standard prediction.

Reproducing the results

```
go run ./examples/bearing_motor
go test ./examples/bearing_motor
python examples/bearing_motor/viz/make_report.py examples/bearing_motor/bearing_motor_report.pdf
```

Sources: Huber, Elektrotechnik und Maschinenbau 76(8), 169 (1959); Milroy, J. Appl. Mech. 34, 525 (1967); Gruenberg, Am. J. Phys. 46, 1213 (1978); Moyssides & Hatzikonstantinou, IEEE Trans. Magn. 26, 1274 (1990) and 33, 4566 (1997); Watson et al., IEE Proc. A 140 (1993), IEE Proc. Sci. Meas. Technol. 146, 83 (1999); Choo, Soong & Abbott, Proc. SPIE 5649, 700 (2005); Marinov and Ogden, Electronics & Wireless World, April 1989; McDonald, "Ball-Bearing Motor", Princeton (2020); Tewari, Universal Principles of Space and Matter (2017), §6.7.

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